

§1.5

The Mayan Numeration System

A base-20 place value system, with one place that uses 18

Today's Goal

After completing this section, you should be able to:

- Read and write numbers using the Mayan numeration system.

LT.2 Numeration Systems

I can convert numbers between various numeration systems including Roman, Egyptian, Mayan, Babylonian, and Hindu-Arabic.

How Many Digits in Base Twenty?

The Babylonians worked in base 60. The Mayans worked in base 20. How many digits do you think they needed?

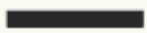
Twenty — the numbers 0 through 19.

And they built all twenty out of just three marks.

Only Three Marks

Every digit from 0 to 19 is built from these three marks.

Table 1.5.2: Mayan Numerals

Mayan Numeral	Number
	0
	1
	5

The shell for zero is believed to be their own invention, appearing about 400 AD.

Twenty Digits from Three Marks

Before you look: a bar is 5 and a dot is 1. How would you build 13?

Table 1.5.3: Mayan Numerals 0 to 19

 = 0	 = 5	 = 10	 = 15
 = 1	 = 6	 = 11	 = 16
 = 2	 = 7	 = 12	 = 17
 = 3	 = 8	 = 13	 = 18
 = 4	 = 9	 = 14	 = 19

Two bars and three dots. Stack the bars, then the dots on top.

Twenty Times the Last Place

The Mayans also used a place value system. However, they used 20 as their base. The value associated with each position in their place value system is a power of 20.

Table 1.5.1: (Pure) Base-20 Place Values

Position	n	...	4th	3rd	2nd	1st
Place Value	20^{n-1}	...	$20^3 = 8,000$	$20^2 = 400$	20	1

While the Babylonians wrote their numerals horizontally, the Mayans chose to write their numerals vertically — highest place value on top.

Reading a Mayan Numeral

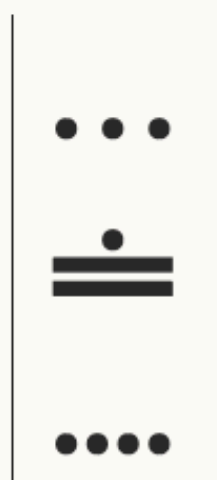
Procedure 1.5.4. Converting a Mayan Numeral to the Corresponding Hindu-Arabic Numeral.

To convert a Mayan numeral to the corresponding Hindu-Arabic numeral, do the following.

1. Identify the place value of each digit in the number.
2. Multiply each digit by its corresponding place value.
3. Write the number as a sum.

Reading a Three-Digit Numeral

What number is represented by this Mayan numeral (assuming a pure base-20 system)?



Three digits, so the places are 400, 20, and 1 — top to bottom.

Solution. Multiply each digit by its place value, then add.

Position	Place Value	Digit	Value of Digit
3rd	400	$\dots = 3$	$3 \times 400 = 1,200$
2nd	20	$\overline{\overline{\cdot}} = 11$	$11 \times 20 = 220$
1st	1	$\dots = 4$	$4 \times 1 = 4$

Adding gives 1,424.

Four Digits, and a Shell

What number is represented by this Mayan numeral (assuming a pure base-20 system)?



Four digits. The shell in the 400 place means that place is empty.

Solution. Multiply each digit by its place value, then add.

Position	Place Value	Digit	Value of Digit
4th	8,000	$\cdot \cdot = 2$	$2 \times 8,000 = 16,000$
3rd	400	 $= 0$	$0 \times 400 = 0$
2nd	20	$\equiv = 16$	$16 \times 20 = 320$
1st	1	$\underline{\cdot} = 7$	$7 \times 1 = 7$

Adding gives 16,327.

Now You Try: Reading

You Try 1.5.1.

What number is represented by this Mayan numeral (pure base-20)?



Position	Place Value	Digit	Value of Digit
3rd	400	 = 4	$4 \times 400 = 1,600$
2nd	20	⊖ = 0	$0 \times 20 = 0$
1st	1	 = 9	$9 \times 1 = 9$
Total. $1,600 + 0 + 9 = 1,609$			

Writing a Mayan Numeral

Procedure 1.5.6. Converting a Hindu-Arabic Numeral to the Corresponding Mayan Numeral.

To convert a Hindu-Arabic numeral to the corresponding Mayan numeral, do the following.

1. If the number is less than 20, then write the corresponding Mayan numeral using Table 1.5.3 and we are done. Otherwise, go to the next step.

Writing a Numeral, continued

Procedure 1.5.6, continued.

2. Determine the highest base-20 place value needed for the number.

Divide the number by this place value.

- a. Use Table 1.5.3 to write the quotient as a Mayan numeral in the corresponding place value.
- b. Return to the first step and use the remainder of this division instead of the original number.

Writing a Number Over 8,000

Write the number 23,486 as a Mayan numeral (pure base-20).

Since $20^3 < 23,486 < 20^4$, we need four digits.

Divide by 8,000.

$$23,486 \div 8,000 = 2 \text{ remainder } 7,486$$

$$7,486 \div 400 = 18 \text{ remainder } 286$$

$$286 \div 20 = 14 \text{ remainder } 6$$

The quotients and the last remainder are our four digits: 2, 18, 14, 6.



Now You Try: Writing

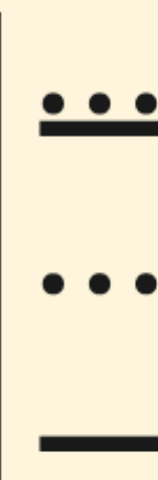
You Try 1.5.2.

Write the number 3,265 as a Mayan numeral (pure base-20).

$$3,265 \div 400 = 8 \text{ remainder } 65$$

$$65 \div 20 = 3 \text{ remainder } 5$$

The three digits are 8, 3, 5.



One Place Breaks the Pattern

The Mayans modified the system for their third place value, using a base-18 instead of 20. That makes the third place $18 \times 20 = 360$ — the length of their 360-day Haab’ year.

Table 1.5.8: Modified Base-20 Place Values

Position	n	...	4th	3rd	2nd	1st
Place Value	$18 \times 20^{n-2}$	...	$18 \times 20^2 = 7,200$	$18 \times 20 = 360$	20	1

Everything above the third place is still multiplied by 20.

The Same Numeral, a New Value

What number is represented by this Mayan numeral (assuming a modified base-20 system)?



This is the numeral from Example 1. In pure base-20 it was 1,424. Watch the top place

Solution. Multiply each digit by its place value, then add.

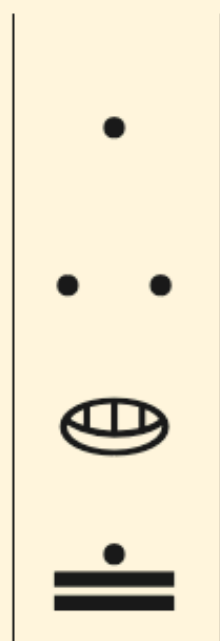
Position	Place Value	Digit	Value of Digit
3rd	360	$\dots = 3$	$3 \times 360 = 1,080$
2nd	20	$\overline{\overline{\cdot}} = 11$	$11 \times 20 = 220$
1st	1	$\dots = 4$	$4 \times 1 = 4$

Adding gives 1,304.

Now You Try: Modified Reading

You Try 1.5.3.

What number is represented by this Mayan numeral (modified base-20)?



Position	Place Value	Digit	Value of Digit
4th	7,200	$\cdot = 1$	$1 \times 7,200 = 7,200$
3rd	360	$\cdot \cdot = 2$	$2 \times 360 = 720$
2nd	20	 $= 0$	$0 \times 20 = 0$
1st	1	$\overline{\overline{\cdot}} = 11$	$11 \times 1 = 11$
Total. $7,200 + 720 + 0 + 11 = 7,931$			

Writing the Modified Way

Write the number 2,678 as a Mayan numeral (modified base-20).

Since $360 < 2,678 < 7,200$, we need three digits. Divide by 360, not 400.

$$2,678 \div 360 = 7 \text{ remainder } 158$$

$$158 \div 20 = 7 \text{ remainder } 18$$

The three digits are 7, 7, 18.



Now You Try: Modified Writing

You Try 1.5.4.

Write the number 25,000 as a Mayan numeral (modified base-20).

$$25,000 \div 7,200 = 3 \text{ remainder } 3,400$$

$$3,400 \div 360 = 9 \text{ remainder } 160$$

$$160 \div 20 = 8 \text{ remainder } 0$$

The four digits are 3, 9, 8, 0.

...

...

...

0

Three Marks, Twenty Digits

- A shell is 0, a dot is 1, a bar is 5. Every digit from 0 to 19 is built from those three marks.
- Numerals are written vertically, highest place value on top.
- Pure base-20 places: 1, 20, 400, 8,000.
- Modified places: 1, 20, 360, 7,200 — the third place uses 18, for the Haab' year.
- To read, multiply each digit by its place value and add. To write, divide by the highest place value and keep going with the remainder.